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Impact of Hip Flexion Angle on Unilateral and Bilateral Nordic Hamstring Exercise Torque and High-Density Electromyography Activity

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Hamstring strain is one of the most common injuries in sports that involve repetitive sprints, such as football,[15] Gaelic football,[35] Australian rules football,[36] and rugby.[8] This type of injury results in significant financial consequences for clubs,[24] as well as substantial player time loss,[22] which are associated with a lower final league ranking, at least in top-level football.[22] Thus, hamstring injury risk reduction is of primary interest in many sports.

SUPPLEMENTAL VIDEO ONLINE

The biarticular hamstrings (biceps femoris long head [BFLH], semitendinosus [ST], and semimembranosus) span both the hip and knee joints, functioning as hip extensors and knee flexors, and play an important role in sprinting.[33] Sprint-type hamstring injuries[2] seem to occur mainly during the late swing phase or early contact phase, in which hamstrings are highly activated at longer lengths.[37] Most sprint-related hamstring injuries occur in the BFLH.[11,26] Although training regimes that aim to induce BFLH fascicle lengthening seem to be beneficial,[40] the injury mechanism is likely to be multifactorial,[6] and other synergist muscles, such as the ST, may play an important role in protecting the BFLH[38] from injury.

In addition to targeting specific muscles, Askling et al[3,4] suggested that for injury prevention, exercises performed at relatively long hamstring muscle lengths may be preferable to those involving short muscle lengths. However, the unilateral exercises in Askling's protocol[39] require relatively low hamstring muscle activity, potentially limiting protective adaptations such as fascicle lengthening and eccentric strength improvements.[6,40]

On the contrary, the Nordic hamstring exercise (NHE) results in higher hamstring muscle activity relative to other exercises.[7,23,42] The NHE has been associated with decreased hamstring injury

BACKGROUND: In the bilateral Nordic hamstring exercise (NHE), hamstrings operate at relatively short lengths, which may limit the efficacy of the NHE in hamstring injury prevention.

OBJECTIVES: To examine knee flexion torque and biceps femoris long head (BFLH) and semitendinosus (ST) high-density electromyography (EMG) activity during the unilateral and bilateral NHE, performed with either neutral (NHE0) or 90°-flexed (NHE90) hips.

METHODS: In this laboratory study, exercises were performed on a novel device at the eccentric 1-repetition maximum load defined for 90° to 15° of knee range of motion. Torque and EMG signals normalized to maximal voluntary isometric activity were compared in different phases of the exercises with statistical parametric mapping.

RESULTS: The EMG levels were lower in NHE90 than in NHE0, mainly in the second half of the movement. Knee flexor eccentric torque was higher in NHE90 than in NHE0 from the beginning to 87% of the bilateral movement, and over the entire unilateral movement. In NHE0, ST activity compared to BFLH activity was higher during the initial movement phase and lower when the movement was close to knee extension. Torque and EMG activity were generally similar in the bilateral and unilateral modes.

CONCLUSION: If performed with neutral hips, the NHE selectively activates the BFLH near full knee extension. Performing the NHE with hips flexed to 90° is preferable when higher passive torque and ST selectivity are targeted at a longer muscle length. Performing these exercises unilaterally could help train each limb separately, with similar torque and EMG output to those of the bilateral conditions. *J Orthop Sports Phys Ther 2019;49(8):584-592. Epub 26 Mar 2019. doi:10.2519/jospt.2019.8801*

KEY WORDS: *biceps femoris, rehabilitation, semitendinosus, strain injury*

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incidence in many studies, provided that the athlete was sufficiently compliant.[6] The NHE is typically performed as an eccentric-only exercise, in which the athlete, in kneeling, performs a slow, bilateral, eccentric action of the knee flexors by resisting knee extension, with neutral or slightly flexed hips. During the NHE, the range of active hamstring lengthening is limited by the neutral hip position, and is further reduced when the athlete is unable to resist the external load at a certain point of knee extension (earlier in weaker individuals). Compared to Askling's protocol,[3,4] the relatively short hamstring operating length during the NHE is considered to be a limitation of this exercise. Although high force and electromyography (EMG) activity during the NHE[23] likely contribute to its preventive effect, it is unknown whether performing the NHE at longer hamstring lengths (ie, with flexed hips) or performing it unilaterally would alter knee flexion torque and hamstring EMG activity during the exercise. This information may help practitioners understand the potential impact of such exercise variations on muscle adaptations.

The primary aim of this study was to examine knee flexion torque and EMG activity of the BFLH and ST during a bilateral NHE performed at a neutral (0°) hip angle (NHE0) with a shorter muscle length, compared to a bilateral NHE performed at 90° of hip flexion (NHE90) with a longer muscle length. The secondary aims were to examine whether knee flexion torque and EMG activity levels differed when performing the exercises in the unilateral compared to the bilateral mode, and to explore intermuscular (BFLH versus ST) differences in EMG activity during the 4 exercises.

METHODS

Participants

Exclusion criteria were history of hamstring or anterior cruciate ligament injury, recent (in the previous 6 months) leg injury, and current musculoskeletal, cardiovascular, or metabolic disorders. Amateur football and rugby players aged 18 to 30 years who trained at least 3 times per week and engaged in strength training on a weekly basis were recruited. Athletes from local sports clubs who met the study criteria and were available to participate volunteered, resulting in a total of 13 participants (mean ± SD age, 23 ± 3 years; body mass, 76 ± 6 kg; height, 1.77 ± 0.05 m). All participants provided written informed consent prior to commencing familiarization with the study protocol. The testing procedures were approved by the Institutional Review Board of the Faculty of Sport Sciences, Université Côte d'Azur (number 2016-08), and performed in an accredited research center (Le Laboratoire Motricité Humaine Expertise, Sport, Santé), according to the Declaration of Helsinki.

Study Design

Participants were familiarized with the study protocol in a 3-week process consisting of 6 sessions to facilitate initial neural adaptations[9] and to take advantage of the "repeated bout effect" on the testing day.[29] During these sessions, a standardized warm-up (**APPENDIX A**, available at www.jospt.org) was followed by practicing all 4 of the test exercises. For each exercise, eccentric action of the knee flexors was performed at a target pace of 5 seconds, controlled with a metronome (Pro Metronome; EUMLab-Xanin Technology GmbH, Berlin, Germany), throughout the 90° to 15° knee range of motion (full knee extension, 0°), as measured with the potentiometer of the load-cell device. Participants initiated the knee extension movement from approximately 110° prior to reaching 90°, from where the 5-second pacing began. Two of the 4 exercises were performed in a neutral hip position throughout the knee extension range (NHE0), and the other 2 were performed in 90° of hip flexion (NHE90), as maintained via the trunk support on the frame of the device (**FIGURE 1**).

Both the NHE0 and NHE90 were performed in a unilateral and a bilateral mode (**FIGURE 1**). To match the relative load across exercises, resistance bands or weight plates were attached to the frame of the device to decrease or increase the external load, respectively (**FIGURE 1**). During each familiarization session, the maximal load at which the eccentric movement could be performed in the knee range of motion of 90° to 15° and at the desired pace was defined and used as a training load (1-repetition maximum). This provided a starting point on the testing day, when the eccentric 1-repetition maximum load for each exercise was redefined. In the last familiarization session, knee flexion maximal voluntary isometric contractions (MVICs) were also practiced to decrease interindividual variability when normalizing EMG signals.

The measurement session was 5 to 8 days after the last familiarization session. During this period, participants were asked to refrain from additional eccentric training. In the main measurement session, after preparation and standardized warm-up, MVICs were performed for EMG normalization purposes. During the MVICs, participants lay prone on the load-cell device, with their shin parallel to the ground and their knees flexed at 30°. Knee joints were aligned with the rotation axis of the frame of the device. Hips and trunk were secured in neutral position. The load cells of the device (1 for each leg) were placed above the proximal border of the lateral malleoli. Participants were instructed to perform forceful knee flexion with their self-reported kicking leg (1 left, 12 right; hereafter referred to as "dominant leg") and to maintain maximum contraction intensity for 2 to 3 seconds.

Three MVICs were performed, with a 1-minute rest. Then, 2 repetitions from each of the 4 exercises were performed in a randomized order, with a 1-minute rest between repetitions and a 4-minute rest between exercises. The target loads (eccentric 1-repetition maximum) were individually defined just before data collection for each exercise. Each

exercise was performed at the 5-second pace practiced before, which translated to approximately 15°/s. Knee flexion force, knee angular displacement, and EMG activity of the BFLH and ST were recorded during MVICs and NHEs. No substantial fatigue was reported throughout the measurement session.

Data Collection

After shaving, abrading, and cleaning the skin, we placed 15-channel, high-density EMG arrays (OT Bioelettronica, Turin, Italy) at a 10-mm interelectrode distance along the midline between the borders of each of the BFLH and ST muscles,[23] guided by brightness-mode ultrasonography (Echo Blaster 128; TELEMED Ltd, Vilnius, Lithuania). Proximal-distal positioning of the electrodes was standardized by aligning channel 8-9 (from distal) with the anatomical middle of the BFLH muscle, defined as the distance between the ischial tuberosity and popliteal fossa. To avoid placing the electrodes over the relatively longer distal tendon of the ST, the array for this muscle was placed relatively more proximally, at 1 cm below the proximal edge of the ST tendinous inscription (located with ultrasonography).

Conductive gel (signa gel; Parker Laboratories, Inc, Fairfield, NJ) was applied to the adhesive foam of the arrays, signal quality was checked during submaximal knee flexion, and the arrays were fixed to the skin using tape. The high-density EMG arrays were connected to a 12-bit analog-digital converter and amplifier (OT Bioelettronica), and digital signals were recorded in OT BioLab Version 3.1 (OT Bioelettronica). A reference electrode strap was placed over the contralateral wrist. Electromyographic signals were amplified by a factor of 1000 and sampled at 2048 Hz. Fifteen EMG channels were recorded in the proximal-distal direction from each of the BFLH and ST simultaneously, covering 15 cm along each muscle.

Force was measured in the dominant leg during MVICs and the eccentric exercises using the load cells of the device (ELAF, 1250 N; TE Connectivity, Schaffhausen, Switzerland), and sagittal plane knee angle was measured with the built-in potentiometer of the device (P4500; Novotechnik, Ostfildern, Germany), both sampled at 1000 Hz. Digitized signals were recorded using a custom interface in LabVIEW (Version 8.5; National Instruments, Austin, TX). The NHE0 and NHE90 showed good to excellent between-session and within-session reliability, as tested previously (**APPENDIX B**, available at www.jospt.org). Torque was calculated using the length of the lever arm, measured as the distance between the femur's lateral epicondyle and the center of the load cells. The OT BioLab (OT Bioelettronica) and LabVIEW (National Instruments) recordings were synchronized with a digital pulse, sent via an external trigger to the load-cell device and EMG system simultaneously.

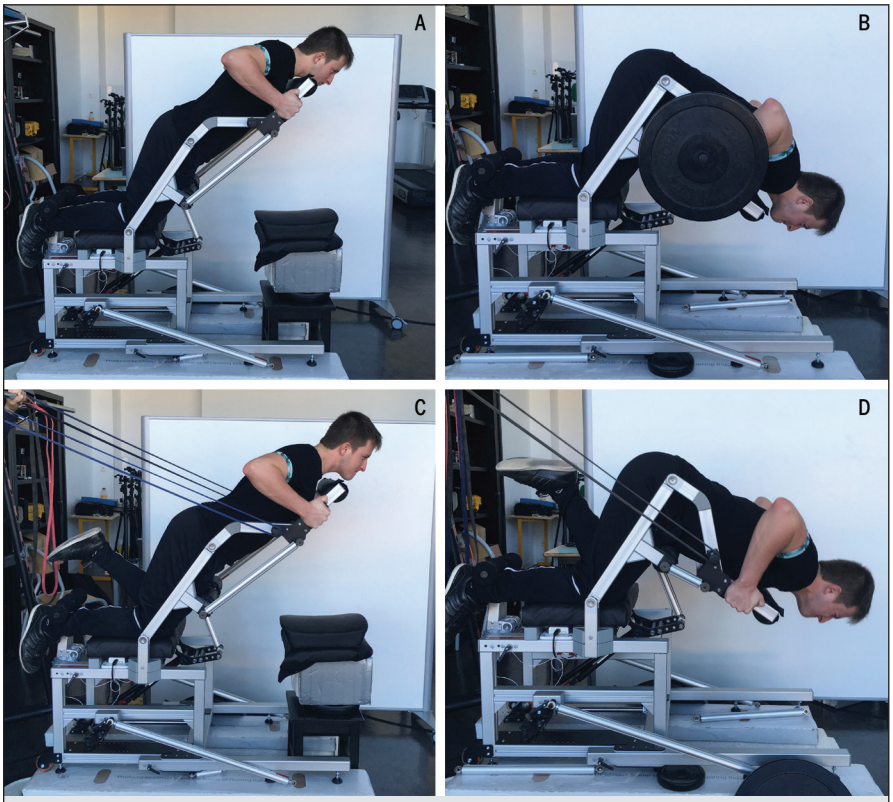

FIGURE 1. The Nordic hamstring exercise was performed with neutral hips (0°; panels A and C) and 90°-flexed hips (panels B and D). The Nordic hamstring exercise with neutral hips required forceful hip extension to minimize hip flexion. In the Nordic hamstring exercise with 90°-flexed hips, the trunk lay on the device, requiring no hip extension effort. Both exercises were performed bilaterally (A and B) and unilaterally (C and D). All exercises were performed at eccentric 1-repetition maximum load, defined for the knee range of motion of 90° to 15° at a pace of 15°/s. Load was adjusted using weight plates or resistance bands, or exercises were performed with body weight only, according to the strength level of the participant.

Data Processing

Only data collected during the active-lengthening phase, standardized across exercises as the range of motion between the knee angles of 90° and 15°, were included in the analyses. Electromyographic signals were band-pass filtered (10-500 Hz) with a zero-lag, fourth-order Butterworth filter in MATLAB (The MathWorks, Inc, Natick, MA). To calculate reference activity levels used for normalization, root-mean-square activity was defined over a 1-second stable period for each EMG channel and MVIC, and for both muscles. Of these MVICs, the highest activity for each channel was used for

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normalization. Normalized signals were smoothed with a 5-Hz, fourth-order, zero-lag, low-pass Butterworth filter. Smoothed curves of all 15 channels per muscle were averaged to estimate overall activity of the BFLH and ST muscles, taking proximal-distal differences[23] into account.

Torque (Newton meters) for the dominant leg (from which EMG signals were recorded) was calculated for each exercise and repetition as the instantaneous force (Newtons) multiplied by the lever arm (meters).

The EMG and torque curves were time normalized (0-100 points) for comparison and included in further analyses as detailed below.

Statistical Analysis

Statistically significant differences between curves were detected using statistical parametric mapping (SPM[20]; Functional Imaging Laboratory, London, UK), by applying 2-tailed paired t tests in MATLAB (The MathWorks, Inc) after ensuring normal data distribution (Shapiro-Wilk test). The SPM analyses were conducted using an open-source code (Version 0.4; www.spm1d.org) in 4 steps. First, the SPM{t} test statistic was calculated for each point in the normalized time series. Then, the critical threshold, which only 5% (alpha level) of smooth random curves would be expected to exceed, was defined. Thereafter, the locations where the SPM{t} trajectory crossed the critical threshold in the normalized time series were determined to define so-called "supra-threshold clusters." Finally, SPM was used to calculate cluster-specific P values. The technical details of SPM are described elsewhere.[1,20]

Using this method, differences in knee flexion torque and BFLH and ST EMG activity were examined between the 2 repetitions of each exercise. This way, we ensured that 2 repetitions per exercise could describe exercise-specific torque and EMG characteristics with high agreement across all points of the normalized time series (**APPENDIX C FIGURES 1** through **3**, available at www.jospt.org). After averaging the 2 repetitions for each exercise, differences between exercises in torque and EMG variables were examined, as well as differences between EMG activity levels of the BFLH and ST for each exercise. Cohen's d_z ± 95% confidence intervals (CIs) were calculated for each point of the normalized time series, presented in **FIGURES 2** through **4** and in **APPENDIX C** (**FIGURES 4** through **8**). Cohen's d_z was calculated as the difference between means divided by the SD of the difference scores. The average Cohen's d_z ± 95% CI for each supra-threshold cluster are presented in the text. Cohen's d_z differences were considered small (0.2 to <0.5), moderate (0.5 to <0.8), or large (≥0.8).[10]

RESULTS

Bilateral NHE0 Versus Bilateral NHE90

In bilateral mode, torque levels in the NHE90 were significantly higher than those in the NHE0 from 0% to 87% of the movement (d_z = 2.74 ± 0.85, P<.001) (**FIGURE 2**).

Both the BFLH and ST exhibited higher EMG activity in the NHE0 than in the NHE90. The BFLH showed 1 supra-threshold cluster from 36% until the end of the movements (d_z = 2.18 ± 1.07, P<.001) (**FIGURE 3**). The ST showed 2 supra-threshold clusters, between 45% and 84% and between 90% and 100% of the movements (d_z = 1.53 ± 1.13, P<.001 and d_z = 1.37 ± 1.13, respectively) (**FIGURE 3**).

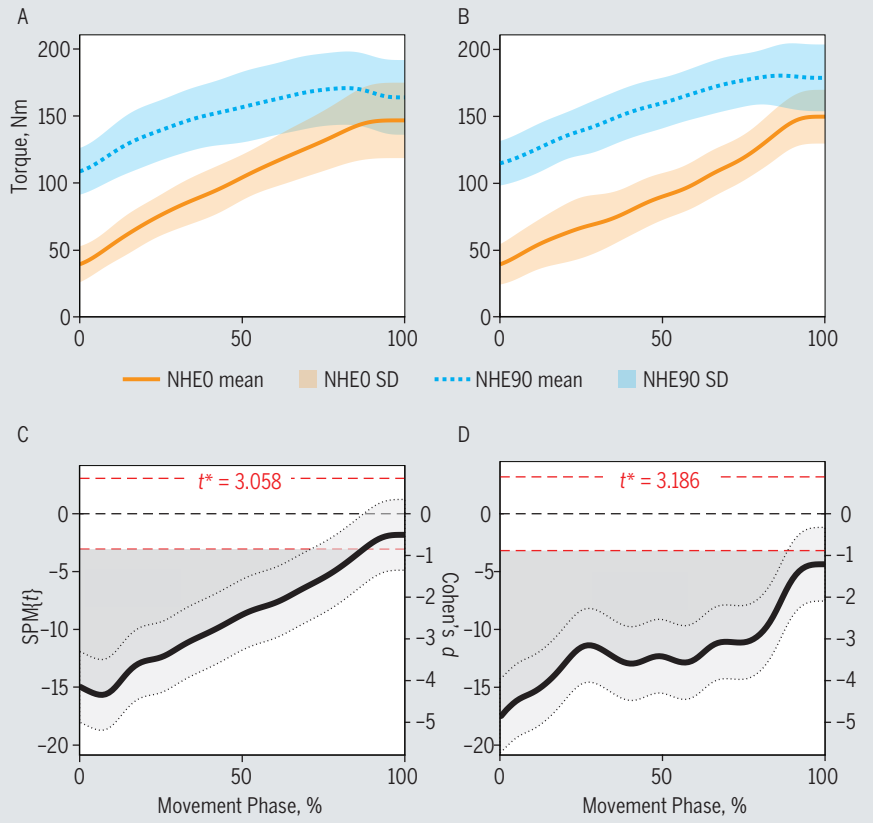

FIGURE 2. Mean and SD values for knee flexion torque in the NHE0 and NHE90, performed (A) bilaterally and (B) unilaterally. Differences between the NHE0 and NHE90 are shown in panels C and D, respectively. Thick black lines represent SPM{t} statistics and Cohen's d effect sizes, and the latter are accompanied by 95% confidence intervals. Critical thresholds (t*) calculated for significance level (α = .05) are shown with dashed red lines. Probability values (P) are calculated for each supra-threshold cluster depicting statistically significant differences (all, P<.001). Abbreviations: NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; SPM, statistical parametric mapping.

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Unilateral NHE0 Versus Unilateral NHE90

In unilateral mode, significantly higher torque levels were found in the NHE90 than in the NHE0 during the entire movement (d_z = 3.22 ± 0.88, P<.001) (**FIGURE 2**).

Both the BFLH and ST exhibited higher activity in the NHE0 than in the NHE90. The BFLH showed 1 supra-threshold cluster from 51% until the end of the movement (d_z = 1.65 ± 1.06, P<.001) (**APPENDIX C FIGURE 4**). The ST showed 3 supra-threshold clusters, at 51% to 53%, 67% to 73%, and 84% to 85% of the movement (d_z = 1.13 ± 1.13, P = .034; d_z = 1.19 ± 1.13, P = .003; and d_z = 1.10 ± 1.13, P = .042, respectively) (**APPENDIX C FIGURE 4**).

Differences Between Bilateral and Unilateral Exercises

The bilateral NHE0 showed higher torque than the unilateral NHE0 in the midrange of the movement (52%-78%, d_z = 0.95 ± 0.91, P = .005) (**APPENDIX C FIGURE 5**). In the NHE90, no difference was found between bilateral and unilateral torques (**APPENDIX C FIGURE 5**).

No difference was found in EMG activity levels in any of the muscles when bilateral exercises were compared to unilateral exercises (**APPENDIX C FIGURES 6** and **7**).

Intermuscular EMG Differences

As shown in **FIGURE 4**, in the bilateral NHE0, ST activity was higher than BFLH activity during the early phase of the movement (2%-4%, d_z = 1.07 ± 1.07, P = .039) and lower during the final phase (94%-98%, d_z = 1.11 ± 1.07, P = .030). Differences between muscles in the unilateral NHE0 did not reach statistical significance (**APPENDIX C FIGURE 8**).

In the bilateral NHE90 (**FIGURE 4**), ST activity, compared to BFLH activity, was statistically higher in 3 subphases in the second half of the movement (62%-69%, d_z = 1.12 ± 1.05, P = .016; 72%-80%, d_z = 1.09 ± 1.05, P = .009; and 83%-100%, d_z = 1.25 ± 1.05, P<.001, respectively). In the unilateral NHE90 (**APPENDIX C FIGURE 8**), higher ST activity, compared to BFLH activity, was observed at the initial and final phases of the movement (0%-3%, d_z = 1.14 ± 1.04, P = .039; 66%-72%, d_z = 1.11 ± 1.04, P = .021; 85%-89%, d_z = 1.10 ± 1.04, P = .034; and 94%-100%, d_z = 1.15 ± 1.04, P = .024, respectively).

DISCUSSION

The present study used a novel load-cell device and high-density EMG to assess knee flexion torque and hamstring EMG output for different NHE modes. Statistical parametric mapping showed remarkably higher torque and considerably lower hamstring EMG activity levels during the NHE90 compared with the NHE0 in most phases of the movements. Muscle activity also differed between the NHE90 and NHE0 in the BFLH relative to the ST. There were similar peak torque and EMG activity levels in unilateral and bilateral modes of both the NHE0 and NHE90, provided that relative loads were matched.

During the bilateral NHE0, high EMG activity was observed in hamstring muscles, a finding similar to that of previous studies involving the NHE.[7,23,42] A recent study used high-density EMG and found higher ST than BFLH activity in the NHE,[23] similar to other studies that have used muscle functional magnetic resonance imaging.[5,7,16,30,31] However, we analyzed data across the entire move-

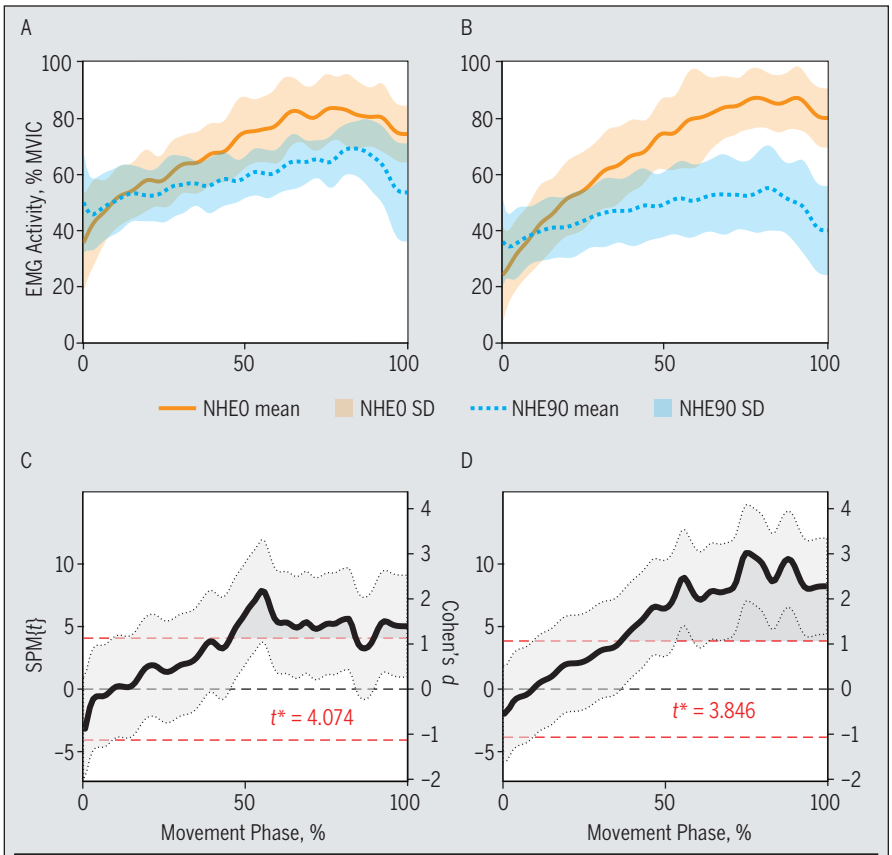

FIGURE 3. Mean and SD values for (A) semitendinosus and (B) biceps femoris long head EMG activity (normalized to percent knee flexion MVIC) in the bilateral exercises (NHE0 and NHE90). Differences between the bilateral NHE0 and NHE90 for each muscle are shown in panels C and D, respectively. Thick black lines represent SPM{t} statistics and Cohen's d effect sizes, and the latter are accompanied by 95% confidence intervals. Critical thresholds (t*) calculated for significance level (α = .05) are shown with dashed red lines. Probability values (P) are calculated for each supra-threshold cluster depicting statistically significant differences (all, P<.001). Abbreviations: EMG, electromyography; MVIC, maximal voluntary isometric contraction; NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; SPM, statistical parametric mapping.

ment and found higher ST than BFLH activity only during the early phase of the movement. The relative activity of the BFLH compared to the ST then increased toward full knee extension. Ultimately, higher BFLH than ST activity reached statistical significance when the knee was relatively extended. Similar results have been reported in isokinetic and isometric knee flexion.[25,34]

In most studies of the NHE, participants performed the exercise at a supramaximal eccentric load.[7,14,23,28,41,42] When doing so, the hamstrings are highly loaded, but participants are generally unable to maintain resistance over the entire range of motion until the knee is nearly extended, at which point the BFLH would be selectively activated according to our results. Another difference between this study and previous studies is that the NHE0 in this study was performed with hip extension effort throughout the movement, with which participants were highly familiarized. The relative activity of the BFLH compared to the ST is higher in hip extension compared to knee flexion exercises.[6] This suggests that hip extension effort in the NHE may improve BFLH activity and might also have contributed to the relatively higher BFLH activity in the current study than that observed in previous studies. From a practical perspective, there seems to be a tradeoff between adding extra load (limiting the range of motion) and selectively activating the BFLH in the NHE0, which should be considered when targeting different muscles in the training program.

In the bilateral NHE90, the hips were flexed to 90° and EMG activity levels of both muscles were substantially lower than those in the NHE0, with the hips in a neutral position. These differences were larger than those reported for eccentric isokinetic testing.[25] In contrast, this difference is not present in isometric knee flexion at different hip flexion angles.[32] This might be explained by eccentric inhibition in the NHE90, where muscle lengths are substantially longer than in the NHE0. In the NHE90, long muscle lengths are accompanied by higher hamstring activity relative to the exercises used in Askling's protocol.[39] Therefore, the NHE90 may be beneficial when aiming to decrease the inhibition associated with hamstring injuries.[21]

Although the protective mechanisms of exercises performed at long muscle lengths are unclear, the NHE90 may be useful when the training goal is to eccentrically activate the hamstrings at long muscle lengths. However, it may also be important that the activity of the BFLH relative to that of the ST differs between the NHE0 and NHE90. Lower BFLH compared to ST activity might be explained by the fact that the NHE90 was performed without hip extension effort (the trunk was supported by the frame of the device). This suggests that performing the NHE with flexed hips might also be beneficial when attempting to selectively target the ST muscle during training.

Despite lower EMG activity levels in the NHE90 than in the NHE0, substantially higher knee flexion torque was observed in the NHE90 than in the NHE0. This agrees with results on isokinetic eccentric action of the knee flexors.[25] Although the difference between knee flexion torques decreased as the knee was extended in the current study, EMG

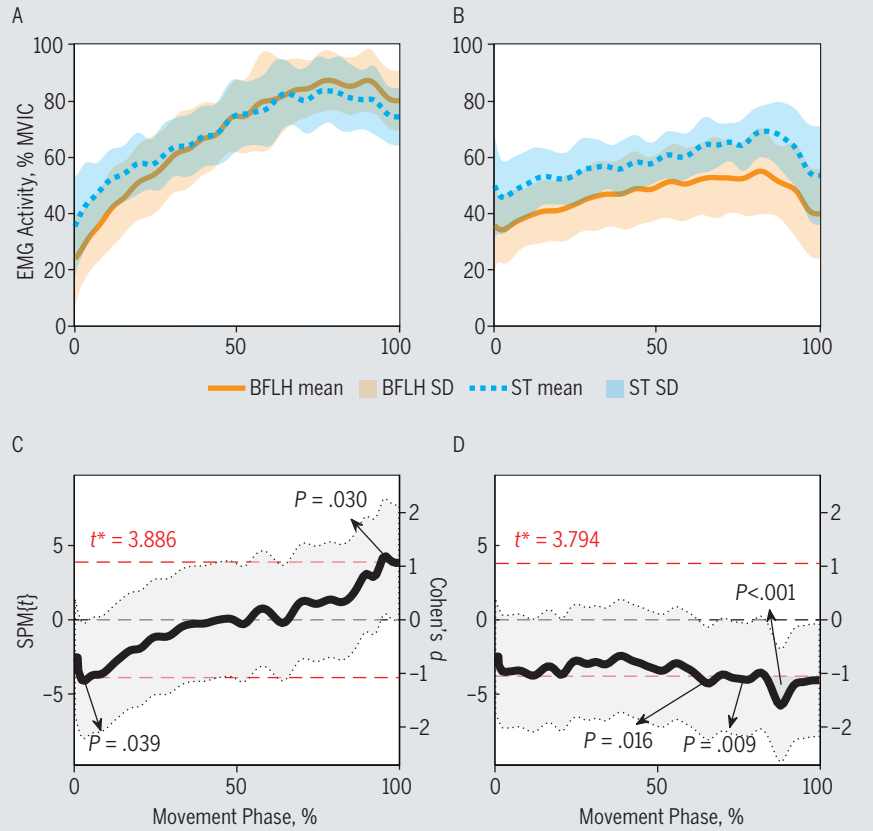

FIGURE 4. Mean and SD values for BFLH and ST EMG activity (normalized to percent knee flexion MVIC) in the bilateral (A) NHE0 and (B) NHE90. Differences between muscles are shown in panels C and D, respectively. Thick black lines represent SPM{*t*} statistics and Cohen's *d* effect sizes, and the latter are accompanied by 95% confidence intervals. Critical thresholds (*t**) calculated for significance level (α = .05) are shown with dashed red lines. Probability values (*P*) are calculated for each supra-threshold cluster depicting statistically significant differences. Abbreviations: BFLH, biceps femoris long head; EMG, electromyography; MVIC, maximal voluntary isometric contraction; NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; SPM, statistical parametric mapping; ST, semitendinosus.

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activity was still lower in the NHE90 close to knee extension. One reason for relatively higher torque in the NHE90 could be that the passive stiffness of hamstring muscles increases with hip flexion.[27] However, the difference in passive stiffness between different hip flexion angles decreases as the knee moves from a flexed toward an extended position,[27] as opposed to what we observed in knee flexion torques. This suggests that passive stiffness has a negligible effect on the substantially higher torque in the NHE90 compared to the NHE0.

Additionally, higher passive stiffness would not explain why EMG activity was so much lower in the NHE90 than in the NHE0. A superior contribution of other knee flexor muscles cannot be totally excluded. However, it seems more likely that high torque coupled with low EMG activity is a result of more pronounced eccentric work at the fascicle level when performing the NHE90 compared to the NHE0, although this hypothesis should be supported experimentally. This, in theory, would further facilitate BFLH fascicle elongation, which seems to be associated with lower risk of hamstring injury.[6,40]

In addition to differences at the fascicle level, it is likely that higher knee flexion torque in the NHE90 compared to the NHE0 also exposes the tendinous structures of the hamstrings to relatively higher forces. Previous studies have shown that a lower ratio between the width of the BFLH proximal aponeurosis and the width of the BFLH muscle results in higher strain at the proximal muscle-tendon junction,[17,18] where most hamstring injuries occur.[13] It should be examined whether exercises with relatively high passive and low active forces—as observed in the NHE90—can increase this ratio, and whether that helps to protect against hamstring injuries.

When comparing bilateral to unilateral exercises, similar torque and EMG levels were reached in the NHE90. Although some differences were found between bilateral and unilateral torques in the NHE0, this was not evident around peak torque. Although the NHE is classically performed bilaterally, unilateral performance may improve the potential to selectively strengthen the weaker and/or injured side without impairing torque and EMG output compared to the bilateral modality. Whether unilateral or bilateral strengthening may be more effective at reducing side-to-side strength imbalances and the risk of hamstring injury warrants further study.[19]

Clinical Implications Summary

The NHE90 may be preferable when selective ST activation and high passive force are targeted at relatively long hamstring operating lengths. Higher torque and lower EMG activity in the NHE90 than in the NHE0 implies (1) more pronounced eccentric work in the NHE90, which may facilitate protective adaptations (eg, fascicle elongation)[39]; and (2) potential for increasing the width ratio of the BFLH proximal aponeurosis and muscle belly, which may reduce injury risk.[14,15]

Performing the NHE with hip extension effort and control over the knee range of motion (applying loads that allow active hamstring lengthening close to knee extension) may be important when more balanced ST and BFLH activation is a training goal.

Performing NHE in unilateral mode may help to decrease interlimb differences in strength and muscle architecture.

Limitations

As a study limitation, surface EMG recordings are prone to cross-talk. In order to limit this effect, we used high-density arrays with small electrodes and a short interelectrode distance,[12] positioned the electrodes with ultrasound guidance, and recruited male athletes who had a relatively thin subcutaneous layer over the hamstrings. Covering 15 cm along the muscles also minimized the effect of muscle movement relative to the skin. The EMG normalization method also affects the observed activity level of these muscles, and it is still unclear whether knee flexion MVIC can activate both the BFLH and ST to a similar extent. Regarding kinematics, knee joint angle was not measured directly but rather with the potentiometer of the device, with some inherent inaccuracy.

A further limitation is the small sample size in this study, which might have led to increased false-negative error rate and biased Cohen's d_z. Although all observed statistically significant differences can be considered large (d_z greater than 1.0), and SPM exerts tight control over type I errors, we encourage replication studies to eliminate potential type I errors in this study. Finally, data obtained from amateur athletes in this study may not be applicable to other populations.

CONCLUSION

We found higher torque and lower EMG activity in the NHE90 compared to the NHE0, and performing these exercises unilaterally elicited similar torque and EMG levels to those in the bilateral exercises. Statistical parametric mapping revealed that controlling the knee range of motion can influence the relative activity of hamstring muscles in the NHE when hips are in a neutral position. These NHE variations may improve the diversity of exercise selection in hamstring-strengthening protocols. Adaptations in response to these exercises, as well as the impact of these exercises on injury prevention, should be further examined. ◉

KEY POINTS

FINDINGS: The Nordic hamstring exercise (NHE) selectively activated the semitendinosus (ST) in the early phase but the biceps femoris long head (BFLH) in the late phase, when hips were in neutral position (0°). On the contrary, performing the NHE with hips flexed to 90° resulted in higher torque and higher activity of the ST compared to the BFLH. The unilateral and bilateral

NHEs resulted in similar knee flexion torques and EMG activity when relative loads were matched.

IMPLICATIONS: The NHE with hips flexed to 90° may be preferable when selective ST activation and high passive force are targeted at relatively long hamstring lengths. Hip extension effort and utilizing the terminal phase of the NHE may be important when more balanced ST and BFLH activation is a training goal.

CAUTION: Adaptations to these variations of the NHE should be further examined before implementing these findings in clinical practice.

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APPENDIX A

DESCRIPTION OF THE STANDARDIZED WARM-UP UTILIZED IN THIS STUDY

1. Five minutes of jogging on a treadmill (self-selected speed: 10-12 km/h)
2. Eight repetitions of "the extender" for each leg
   The participant lies supine, with the hip flexed at approximately 90° on 1 side and the other leg in neutral position. The participant stabilizes the thigh over the flexed hip with his or her hands and performs slow, cyclic knee flexion/extension in the full knee range of motion
3. Eight repetitions of "the diver" for each leg
   Standing on 1 leg (knee flexion, approximately 10°), the participant performs a simulated dive, that is, flexes the hip while lifting the other leg (similar to a 1-leg Romanian deadlift but with a 90° knee angle), and simultaneously extends the arms forward. The pelvis is kept horizontal and knee angles are constant during the exercise. After reaching maximal hip flexion, the participant returns to the starting position
4. Two repetitions of the traditional Nordic hamstring exercise, performed on the ground while the ankles are stabilized manually
5. Repeat steps 2 through 4
6. One maximal isometric contraction on the device, performed with neutral trunk and hips and the knee flexed to 30° (same position and instructions as when recording maximal voluntary isometric contractions for normalization purposes)

"The extender," "the diver," and the Nordic hamstring exercise are presented in the **ONLINE VIDEO**.

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APPENDIX B

RELIABILITY OF THE LOAD-CELL DEVICE USED IN THIS STUDY

Participants

Prior to data collection in this study, the reliability of peak force measured by the device was tested on 20 male amateur athletes without prior hamstring injury (mean ± SD age, 21.3 ± 2.9 years; height, 1.78 ± 0.05 m; body mass, 75.8 ± 8.4 kg). All participants gave written consent to participate in this reliability study. Data collection was conducted in an accredited laboratory (Le Laboratoire Motricité Humaine Expertise, Sport, Santé, Nice, France), and study procedures were approved by the Faculty of Sport Sciences at the Université Côte d'Azur (number 2016-08) and done according to the Declaration of Helsinki.

Study Design

Participants performed the bilateral Nordic hamstring exercise with neutral hips (0°) and with hips flexed to 90°. A metronome was used to control the pace of each exercise at 15°/s, to minimize differences between repetitions, exercises, and participants. All exercises were performed with body weight only. Peak force was defined for the left and right legs. In each of the 2 sessions, 2 repetitions of each exercise were performed in a random order after warm-up. There were 7 to 10 days of rest between the 2 sessions. Within-session reliability results reflect the reliability of the 2 repetitions performed per session (**TABLE 1**). Between-session reliability is based on the average of the 2 repetitions per session (**TABLE 2**).

Calculated Reliability Measures

The intrasubject coefficient of variation (CV) was calculated as CV % = $\frac{SDi}{\bar{z}_i} \times 100$, where SD*i* is the standard deviation of the 2 repetitions and $\bar{z}_i$ is the mean of the 2 repetitions in each participant. The tables below show group averages, with 95% confidence intervals.

Typical error (TE) = $\frac{SD\ (y_i - x_i)}{\sqrt{2}}$, where SD is the standard deviation of the difference between y_i and x_i observations. Typical error is also expressed as a CV (TE %; percentage of the mean).

The intraclass correlation coefficient (model 3,1) $r = 1 - \frac{TE^2}{SD^2}$, where TE is the typical error and SD is the mean between-subject standard deviation.

Small-sample bias was tested, based on a factor calculated as $1 + \frac{1 - r^2}{2(n-3)}$, where n is the sample size and *r* is the intraclass correlation coefficient.

Minimum detectable change at the 95% confidence level = TE × 1.96 × $\sqrt{2}$, where TE is the typical error.

TABLE 1
Within-Session Reliability

	NHE0		NHE90	
	Left Leg	Right Leg	Left Leg	Right Leg
Session 1*				
Repetition 1	380.1 ± 63.2	375.7 ± 63.7	354.0 ± 46.7	345.6 ± 61.0
Repetition 2	367.7 ± 63.1	367.6 ± 68.5	358.1 ± 42.9	365.4 ± 56.5
Session 2*				
Repetition 1	388.4 ± 54.9	376.8 ± 53.6	364.5 ± 54.6	322.0 ± 52.6
Repetition 2	385.6 ± 62.5	374.2 ± 53.9	358.5 ± 46.3	330.6 ± 53.2
ICC†				
Session 1	0.93 (0.82, 0.97)	0.90 (0.77, 0.96)	0.94 (0.85, 0.97)	0.94 (0.87, 0.98)
Session 2	0.94 (0.85, 0.97)	0.87 (0.70, 0.95)	0.91 (0.78, 0.96)	0.95 (0.88, 0.98)
CV, %†				
Session 1	4.40 (3.07, 5.74)	4.74 (3.28, 6.21)	3.17 (2.39, 3.96)	4.43 (2.11, 6.75)
Session 2	3.25 (2.10, 4.39)	4.28 (2.65, 5.92)	3.19 (1.97, 4.42)	3.36 (2.07, 4.64)

Table continues on page B3.

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APPENDIX B

	NHE0		NHE90	
	Left Leg	Right Leg	Left Leg	Right Leg
TE†				
Session 1	18.1 (13.8, 26.5)	21.7 (16.5, 31.7)	12.0 (9.1, 17.5)	14.6 (11.1, 21.3)
Session 2	15.5 (11.8, 22.6)	20.3 (15.4, 29.7)	16.2 (12.3, 23.7)	12.3 (9.2, 18.4)
TE, % of mean†				
Session 1	4.85 (3.69, 7.08)	5.84 (4.44, 8.54)	3.37 (2.56, 4.92)	4.11 (3.12, 6.00)
Session 2	4.00 (3.04, 5.85)	5.41 (4.11, 7.90)	4.49 (3.41, 6.56)	3.77 (2.83, 5.65)
MDC_{95}				
Session 1	50.2	60.2	33.2	40.5
Session 2	42.9	56.3	45.0	34.1

Abbreviations: CV, intrasubject coefficient of variation; ICC, intraclass correlation coefficient; MDC_{95}, minimum detectable change at the 95% confidence level; NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; TE, typical error.
**Values are mean ± SD peak force, measured in each leg, in the NHE0 and NHE90.*
†Values in parentheses are 95% confidence interval.

TABLE 2
Between-Session Reliability

	NHE0		NHE90	
	Left Leg	Right Leg	Left Leg	Right Leg
Session 1*	373.9 ± 61.9	371.6 ± 64.3	356.1 ± 44.1	351.3 ± 56.2
Session 2*	387.0 ± 57.8	375.5 ± 51.8	361.5 ± 49.3	326.3 ± 52.2
ICC†	0.91 (0.78, 0.96)	0.81 (0.59, 0.92)	0.89 (0.75, 0.96)	0.85 (0.65, 0.94)
CV, %†	4.38 (2.87, 5.90)	5.35 (3.31, 8.88)	3.86 (2.88, 4.84)	7.10 (5.46, 8.75)
TE†	19.4 (14.7, 28.23)	26.3 (20.0, 38.5)	16.2 (12.3, 23.6)	22.1 (16.7, 32.7)
TE, % of mean†	5.09 (3.87, 7.42)	7.05 (5.36, 10.30)	4.51 (3.43, 6.58)	6.53 (4.94, 9.66)
MDC_{95}	53.6	73.0	44.8	61.3

Abbreviations: CV, intrasubject coefficient of variation; ICC, intraclass correlation coefficient; MDC_{95}, minimum detectable change at the 95% confidence level; NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; TE, typical error.
**Values are mean ± SD peak force, measured in each leg, in the NHE0 and NHE90. In session 2, right-leg force in the NHE90 from 1 participant was excluded due to data loss when exporting; therefore, in this comparison—as well as when the NHE90 right-leg force was compared between sessions—we had 19 instead of 20 observations.*
†Values in parentheses are 95% confidence interval.

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APPENDIX C

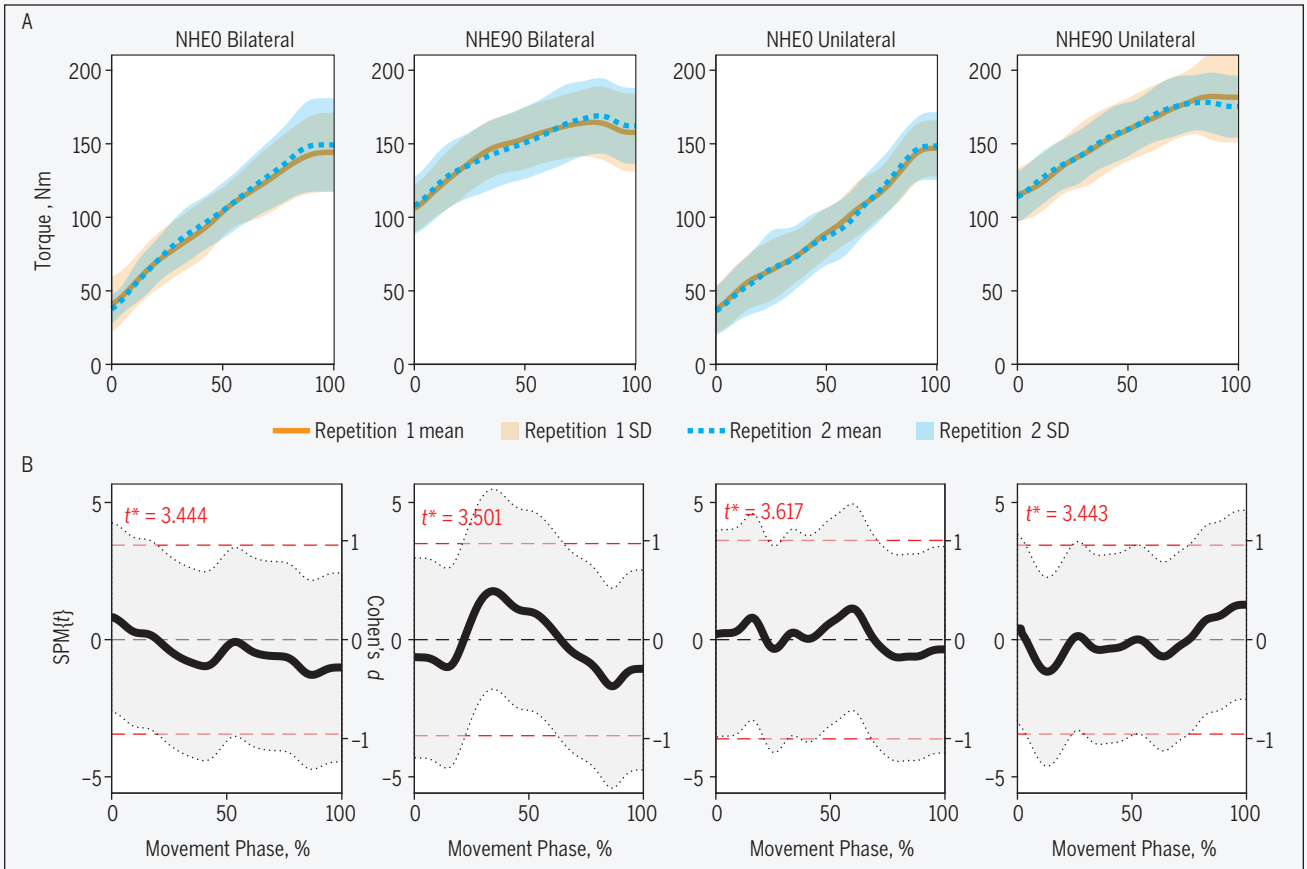

FIGURE 1. Torque differences between the first and second repetitions. (A) Mean and SD values for knee flexion torque for each exercise. (B) Differences between repetitions for each exercise. Thick black lines represent SPM{*t*} statistics and Cohen's *d* effect sizes, and the latter are accompanied by 95% confidence intervals. Critical thresholds (*t**) calculated for significance level (α = .05) are shown with dashed red lines. No statistically significant differences were found between repetitions. Abbreviations: NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; SPM, statistical parametric mapping.

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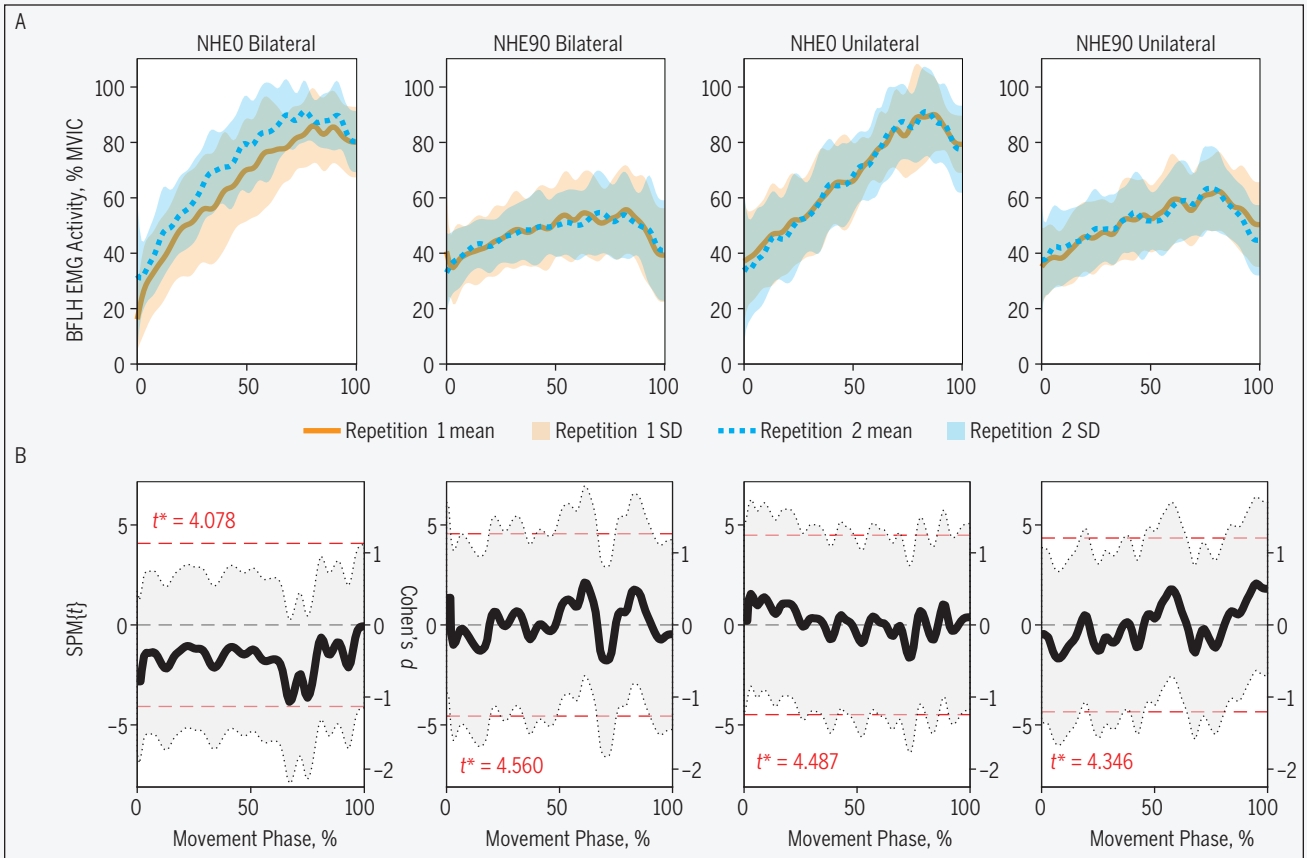

FIGURE 2. Differences for the BFLH between the first and second repetitions. (A) Mean and SD values for BFLH EMG activity for each exercise when normalized to percent MVIC. (B) Differences between repetitions for each exercise. Thick black lines represent SPM{*t*} statistics and Cohen's *d* effect sizes, and the latter are accompanied by 95% confidence intervals. Critical thresholds (*t**) calculated for significance level (α = .05) are shown with dashed red lines. No statistically significant differences were found between repetitions. Abbreviations: BFLH, biceps femoris long head; EMG, electromyography; MVIC, maximal voluntary isometric contraction; NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; SPM, statistical parametric mapping.

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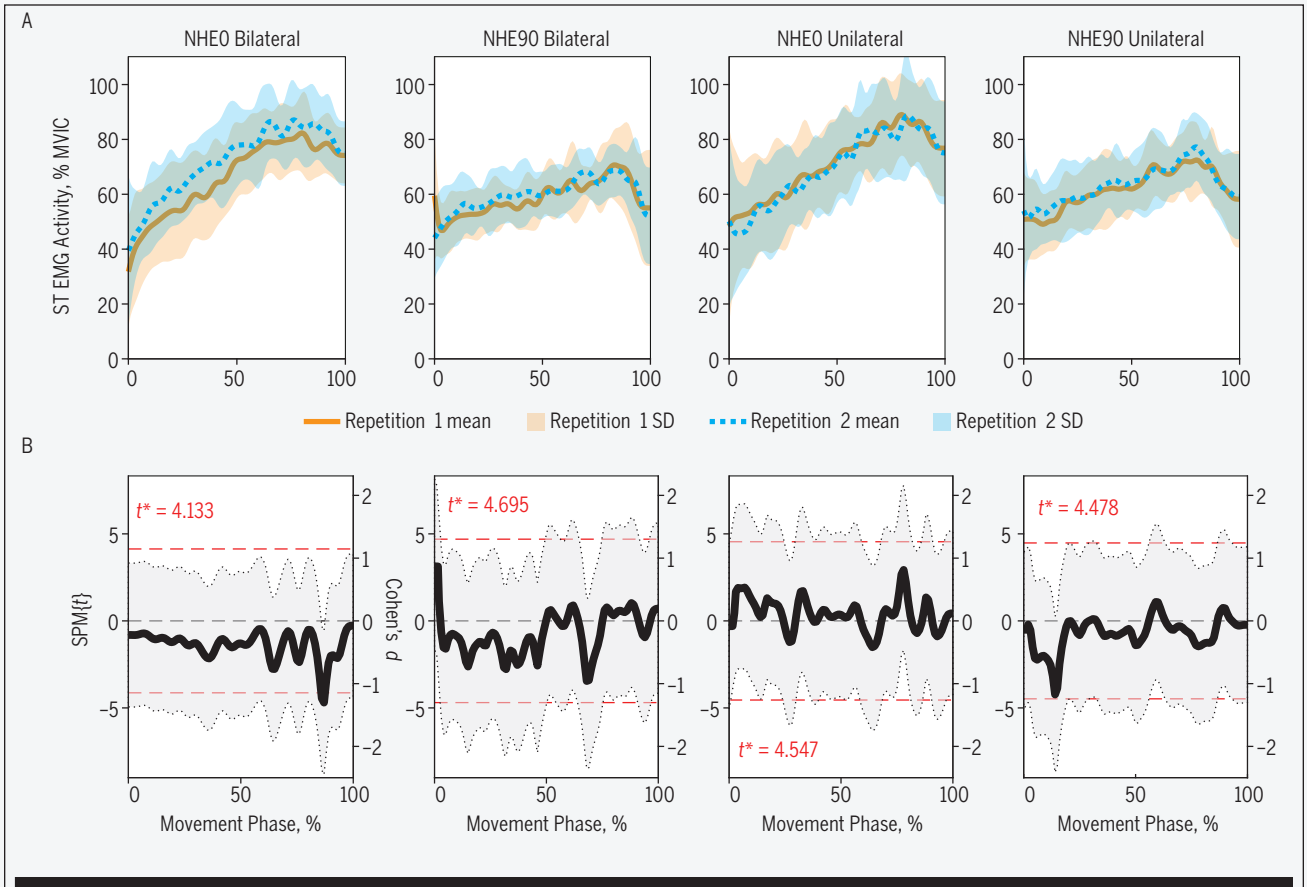

FIGURE 3. Differences for the ST between the first and second repetitions. (A) Mean and SD values for ST EMG activity for each exercise when normalized to percent MVIC. (B) Differences between repetitions for each exercise. Thick black lines represent SPM{*t*} statistics and Cohen's *d* effect sizes, and the latter are accompanied by 95% confidence intervals. Critical thresholds (*t**) calculated for significance level (α = .05) are shown with dashed red lines. The only statistically significant difference was found in the bilateral NHE0, between 85% and 86% of the movement phase (*P* = .038). Abbreviations: EMG, electromyography; MVIC, maximal voluntary isometric contraction; NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; SPM, statistical parametric mapping; ST, semitendinosus.

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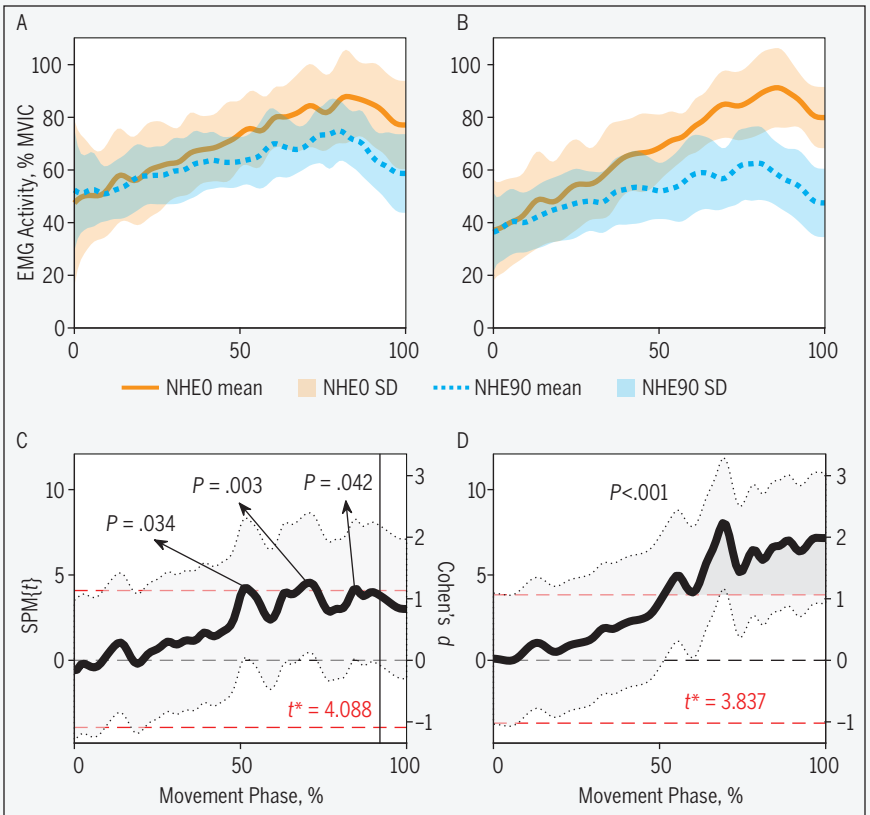

FIGURE 4. Differences between the unilateral NHE0 and NHE90 for each muscle. Mean and SD values for (A) semitendinosus and (B) biceps femoris long head EMG activity for each exercise when normalized to percent MVIC. Differences between (C) the unilateral NHE0 and (D) the unilateral NHE90 for each muscle. Thick black lines represent SPM{*t*} statistics and Cohen's *d* effect sizes, and the latter are accompanied by 95% confidence intervals. Critical thresholds (t^*) calculated for significance level ($\alpha = .05$) are shown with dashed red lines. Probability values (*P*) are calculated for each supra-threshold cluster depicting statistically significant differences. Abbreviations: EMG, electromyography; MVIC, maximal voluntary isometric contraction; NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; SPM, statistical parametric mapping.

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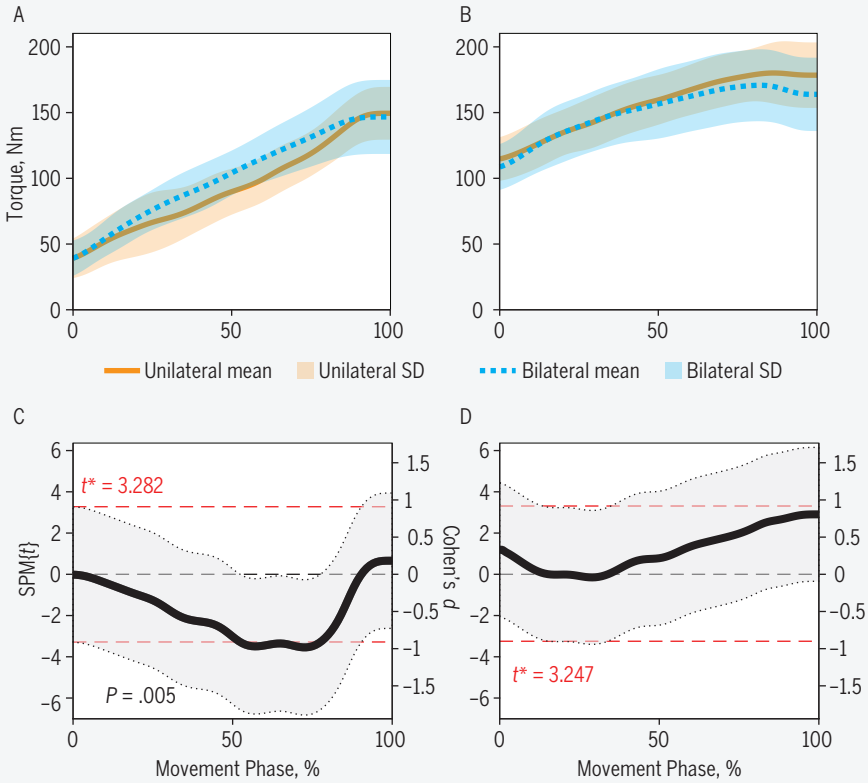

FIGURE 5. Differences between unilateral and bilateral torque. Mean and SD values for knee flexion torque in (A) the bilateral and unilateral NHE0 and (B) the bilateral and unilateral NHE90. Differences between (C) unilateral and (D) bilateral exercises. Thick black lines represent SPM{*t*} statistics and Cohen's *d* effect sizes, and the latter are accompanied by 95% confidence intervals. Critical thresholds (*t**) calculated for significance level (α = .05) are shown with dashed red lines. Probability values (*P*) are calculated for each supra-threshold cluster depicting statistically significant differences. Abbreviations: NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; SPM, statistical parametric mapping.

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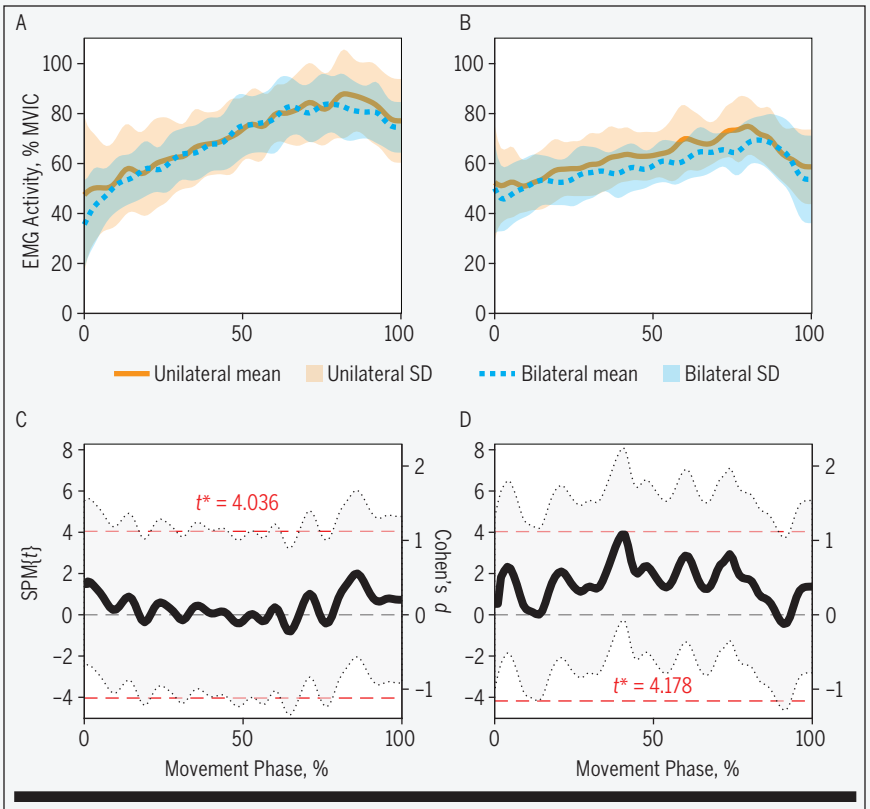

FIGURE 6. Differences in semitendinosus EMG activity between unilateral and bilateral modes of each exercise. Mean and SD values for semitendinosus EMG activity, when normalized to percent MVIC, in the (A) NHE0 and (B) NHE90. Differences between the unilateral and bilateral (C) NHE0 and (D) NHE90. Thick black lines represent SPM{*t*} statistics and Cohen's *d* effect sizes, and the latter are accompanied by 95% confidence intervals. Critical thresholds (t^*) calculated for significance level ($\alpha = .05$) are shown with dashed red lines. Probability values (*P*) are calculated for each supra-threshold cluster depicting statistically significant differences. Abbreviations: EMG, electromyography; MVIC, maximal voluntary isometric contraction; NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; SPM, statistical parametric mapping.

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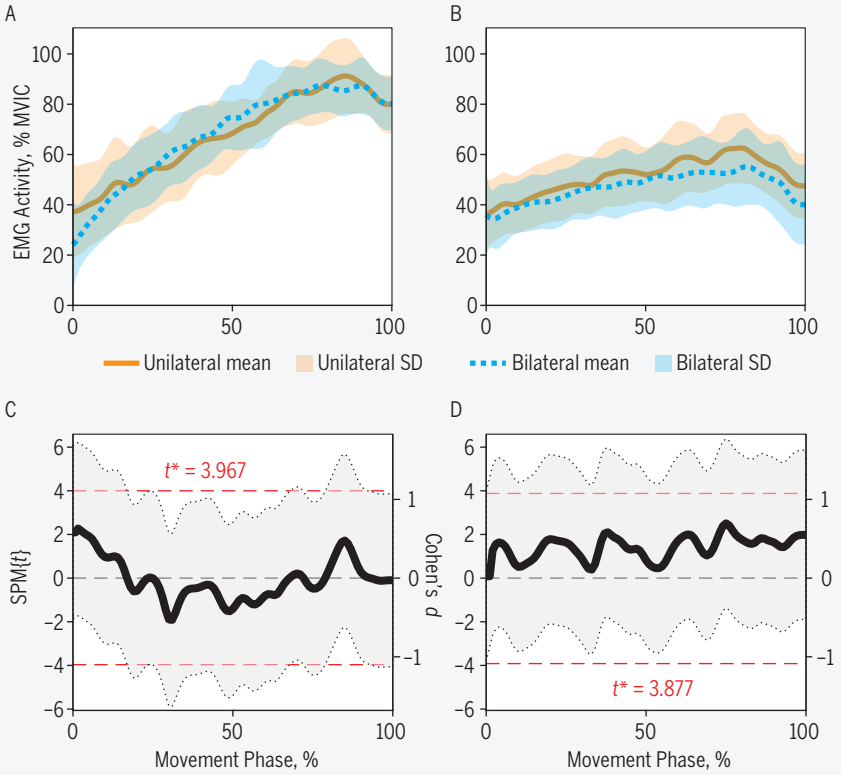

FIGURE 7. Differences in biceps femoris long head EMG activity between unilateral and bilateral modes of each exercise. Mean and SD values for biceps femoris long head EMG activity, when normalized to percent MVIC, in the (A) NHE0 and (B) NHE90. Differences between the unilateral and bilateral (C) NHE0 and (D) NHE90. Thick black lines represent SPM{*t*} statistics and Cohen's *d* effect sizes, and the latter are accompanied by 95% confidence intervals. Critical thresholds (*t**) calculated for significance level (α = .05) are shown with dashed red lines. Probability values (*P*) are calculated for each supra-threshold cluster depicting statistically significant differences. Abbreviations: EMG, electromyography; MVIC, maximal voluntary isometric contraction; NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; SPM, statistical parametric mapping.

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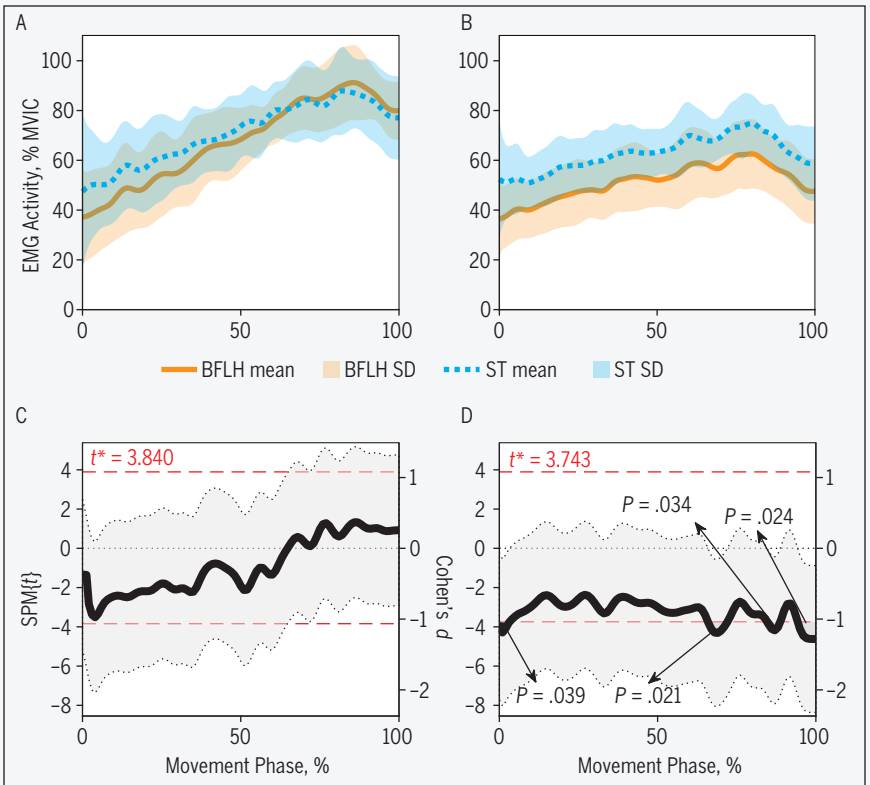

FIGURE 8. Intermuscular differences in the unilateral exercises. Mean and SD values for BFLH and ST EMG activity when normalized to percent MVIC in the (A) NHE0 and (B) NHE90. Differences between muscles are shown in panels C and D, respectively. Thick black lines represent SPM{*t*} statistics and Cohen's *d* effect sizes, and the latter are accompanied by 95% confidence intervals. Critical thresholds (*t**) calculated for significance level (α = .05) are shown with dashed red lines. Probability values (*P*) are calculated for each supra-threshold cluster depicting statistically significant differences. Abbreviations: BFLH, biceps femoris long head; EMG, electromyography; MVIC, maximal voluntary isometric contraction; NHE0, Nordic hamstring exercise with neutral hips; NHE90, Nordic hamstring exercise with 90°-flexed hips; SPM, statistical parametric mapping; ST, semitendinosus.

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